



EduVision_DV (Higher Education Performance Dashboard)

PROJECT DOCUMENTATION

Submitted By

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Project Overview

EduVision DV – Higher Education Performance Dashboard is a data analytics and visualization project designed to analyze and compare the performance of universities using higher-education ranking and institutional data.

The project combines data cleaning, analysis, KPI development, and interactive dashboard visualization to provide meaningful insights into university performance. The dashboard analyzes important areas such as overall university scores, research performance, student population, internationalization, regions, and countries.

The project consists of four main interactive dashboards:

1. **University Overview** – Provides a high-level view of university performance using KPIs and regional comparisons.
2. **Research Analytics** – Analyzes research quality, research environment, teaching, and industry-related performance.
3. **Student Analytics** – Focuses on total student population, international students, student diversity, and regional student trends.
4. **Country Comparison** – Compares universities across countries using overall scores, research quality, student population, maps, pie charts, and other visualizations.

The project is developed using Python for data preparation and Tableau for interactive data visualization and dashboard development. The final dashboard suite is designed to support easy comparison and help users understand patterns and trends in higher education performance.

Project Objectives

The main objective of **EduVision DV – Higher Education Performance Dashboard** is to transform university and higher education data into meaningful and easy-to-understand visual insights.

The key objectives of the project are:

1. **Data Collection and Preparation:** To collect university ranking and higher education data and prepare it for analysis by cleaning and organizing the dataset using Python.
2. **Performance Analysis:** To analyze university performance based on factors such as overall score, research quality, teaching, student population, and internationalization.
3. **KPI Development:** To create meaningful Key Performance Indicators such as total universities, average overall score, research quality, total students, and international student percentage.
4. **Interactive Dashboard Development:** To design interactive dashboards using Tableau that allow users to explore higher education data easily.
5. **University and Country Comparison:** To compare universities, countries, and regions based on different performance measures and identify important differences.
6. **Effective Data Visualization:** To represent the analyzed data using different charts such as **bar charts, column charts, pie charts, maps, scatter plots, and tree maps**.
7. **User-Friendly Analysis:** To provide filters and interactive features that help users view specific regions, countries, and university categories according to their requirements.
8. **Data-Driven Insights:** To identify useful patterns and relationships in higher education data and present them in a clear and understandable manner.

Dataset Sources & Description

Dataset Source

The dataset used for the **EduVision DV – Higher Education Performance Dashboard** was collected from Kaggle, a popular online platform for datasets and data science projects. The dataset contains information related to universities, rankings, academic performance, research, students, and internationalization.

The collected dataset was then cleaned and prepared using Python before importing it into Tableau for analysis and visualization.

Dataset Description

The dataset contains information about **200 universities** and includes various attributes related to university performance. Important fields include:

- **University Name** – Name of the university.
- **Rank and Previous Rank** – Current and previous university rankings.
- **Country/Territory** – Country where the university is located.
- **Region** – Geographical region of the university.
- **Overall Score** – Overall performance score of the university.
- **Research Quality** – Measures the quality of research performance.
- **Research Environment** – Represents the research environment of the university.
- **Teaching Score** – Indicates teaching-related performance.
- **Total Students** – Total number of students in the university.
- **International Students Percentage** – Percentage of international students.
- **Faculty Student Ratio** – Information related to faculty and student proportions.
- **International Student Diversity** – Measures the diversity of international students.
- **University Type** – Classification of the university.
- **Founded** – Year in which the university was established.

The dataset provides a useful foundation for comparing **universities, countries, and regions** and analyzing different aspects of higher education performance.

Data Cleaning & Preparation

The collected dataset was cleaned and prepared before using it for dashboard development. **Python** was used to perform the data cleaning and transformation process.

The main steps followed were:

1. **Data Loading:** The Kaggle dataset was imported into Python for analysis and preprocessing.
2. **Data Inspection:** The dataset was checked to understand the columns, data types, number of records, and missing values.
3. **Handling Missing Values:** Missing values were identified and handled appropriately based on the requirements of each field.
4. **Removing Unnecessary Data:** Unnecessary or duplicate information was removed to make the dataset more organized and suitable for analysis.
5. **Column Name Cleaning:** Column names were standardized to make them easier to use during analysis and visualization.
6. **Data Type Correction:** Numerical, text, and other fields were converted into suitable data types wherever required.
7. **Data Transformation:** Required fields were created or transformed to support KPI calculations and dashboard visualizations.
8. **Final Dataset Preparation:** After cleaning and transformation, the prepared dataset was saved as **Cleaned dataset.csv** and imported into **Tableau** for creating the final dashboards.

This cleaning and preparation process helped improve the accuracy, consistency, and usability of the dataset for further analysis.

KPI Definitions

KPIs (Key Performance Indicators) are important measures used in the **EduVision DV – Higher Education Performance Dashboard**. They provide a quick overview of university performance and help users compare different aspects of higher education.

University Overview

KPI	Definition
Total Universities	Represents the total number of unique universities included in the dataset. It provides an overall view of the number of institutions considered for the analysis.
Average Overall Score	Represents the average overall performance score of all universities. It provides a general indication of the academic and institutional performance of the universities.
Average Research Quality	Represents the average research quality score of the universities. It helps understand the overall strength and quality of research activities.
Average International Students %	Represents the average percentage of international students across the universities. It indicates the level of internationalization and diversity of the institutions.

Research Analytics

KPI	Definition
Average Research Quality	Shows the average research quality score of the universities. It helps evaluate and compare the research performance of different universities and regions.
Average Research Environment	Represents the average research environment score. It provides an understanding of the conditions and academic environment supporting research activities.
Average Teaching Score	Represents the average teaching score of the universities. It helps measure the overall teaching performance and quality across institutions.
Average Industry Score	Shows the average industry-related score of the universities. It provides an indication of their performance and connection with industry-related activities.

Student Analytics

KPI	Definition
Average International Students %	Represents the average percentage of international students across universities. It helps measure the international presence and diversity of the student population.
Total Students	Represents the total number of students across all universities in the dataset. It provides an overall view of the student population being analyzed.
Average Faculty Student Ratio Score	Represents the average faculty-student ratio score. It helps understand the relationship between faculty availability and the student population.
Average International Student Diversity	Represents the average international student diversity score. It helps evaluate the level of diversity among international students in the universities.

Country Comparison

KPI / Measure	Definition
Average Overall Score	Represents the average overall university score for each country. It allows users to compare the general performance of universities across different countries.
Total Students	Represents the total student population associated with universities in each country. It helps identify countries with larger university student populations.
Average Research Quality	Represents the average research quality score of universities within each country. It helps compare research performance between countries.
International Students %	Represents the percentage of international students and is used to analyze the relationship between internationalization and overall university performance.

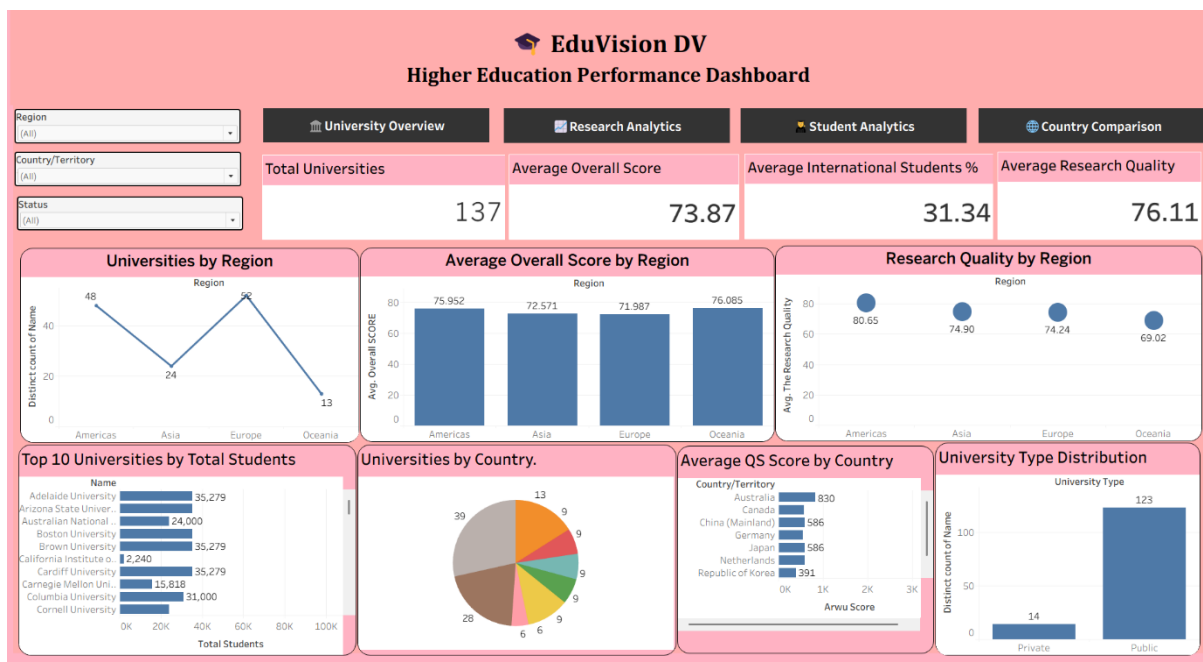
Dashboard Guide

The **EduVision DV – Higher Education Performance Dashboard** consists of four interactive dashboards. Each dashboard focuses on a different aspect of university performance and provides users with clear visual insights through KPIs, charts, maps, filters, and comparisons.

University Overview

The University Overview dashboard provides a high-level summary of the universities included in the dataset. It displays important KPIs such as Total Universities, Average Overall Score, Average Research Quality, and Average International Students %.

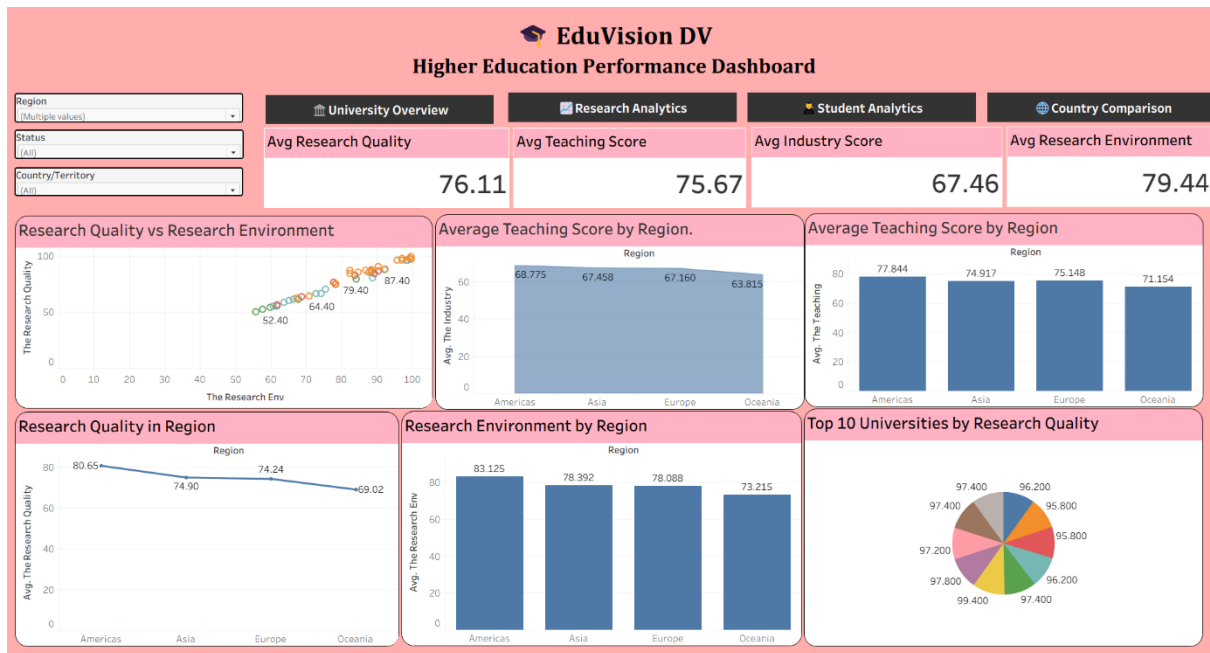
The dashboard includes visualizations such as universities by region, research quality by region, universities by status, top universities, and university-type comparisons. Users can use filters such as Region, Country/Territory, and Status to explore specific universities and categories.



Research Analytics

The Research Analytics dashboard focuses on the research and academic performance of universities. It includes KPIs such as Average Research Quality, Average Research Environment, Average Teaching Score, and Average Industry Score.

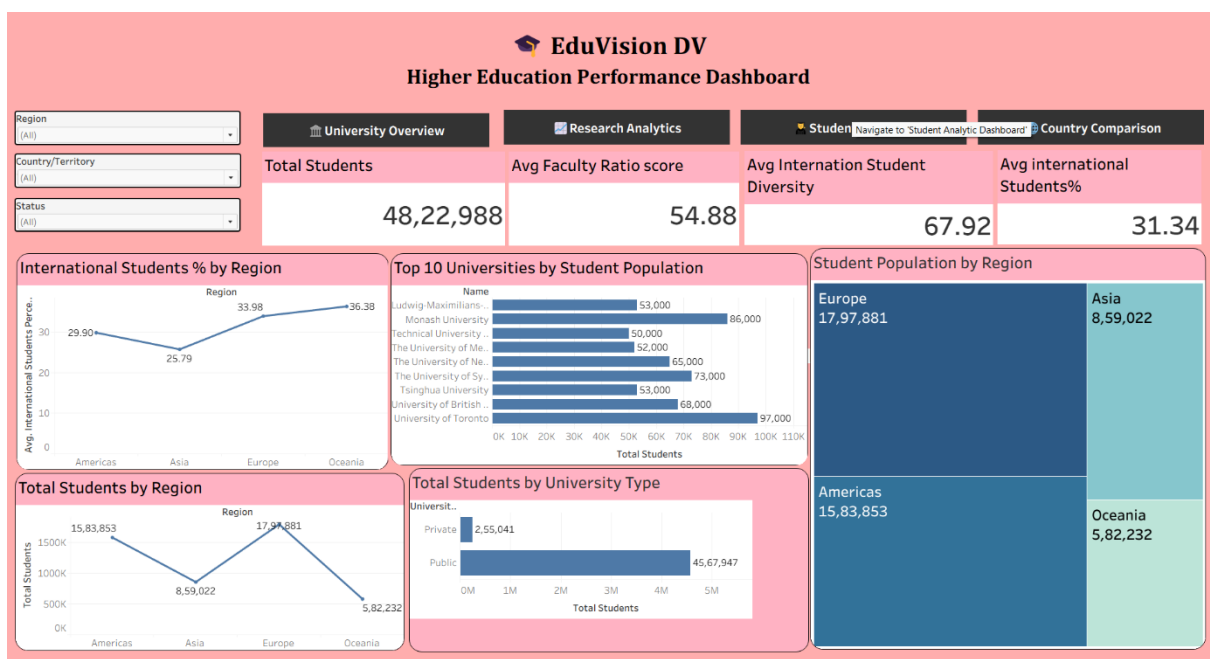
The dashboard uses charts to compare research performance across regions and universities. A scatter plot is also used to understand the relationship between Research Quality and Overall Score. Users can apply filters to analyze research performance for specific regions, countries, or university statuses.



Student Analytics

The Student Analytics dashboard focuses on student population and internationalization. Its main KPIs include Average International Students %, Total Students, Average Faculty Student Ratio Score, and Average International Student Diversity.

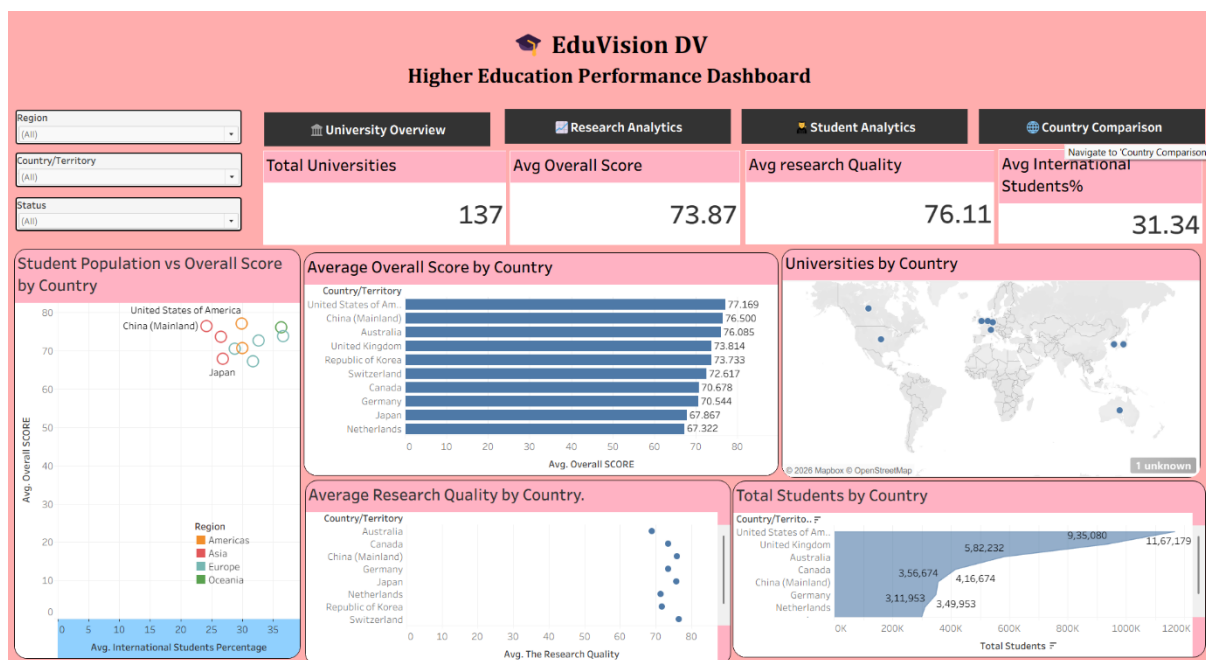
The dashboard contains visualizations such as International Students % by Region, Total Students by Region, Top 10 Universities by Student Population, Student Population Tree Map by Region, and International Students % vs Overall Score. These visualizations help users understand student population patterns and international participation.



Country Comparison

The Country Comparison dashboard is designed to compare university performance across different countries. It uses a geographical map to display universities by country and several charts for comparison.

The dashboard includes Average Overall Score by Country, Total Students by Country, Average Research Quality by Country, Universities by Country pie chart, and International Students % vs Overall Score scatter plot. These visualizations help users identify differences in university performance, student population, research quality, and internationalization between countries.



Interactive Features

All four dashboards provide interactive features such as **Region, Country/Territory, and Status filters**. When a filter is selected, the relevant charts and KPIs update automatically. This allows users to explore the data from different perspectives and obtain more specific insights.

Overall, the dashboard guide helps users understand the purpose of each dashboard and how to use the available **KPIs, visualizations, filters, and comparisons** to analyze higher education performance effectively.

Education Analytics Methodology

The **EduVision DV – Higher Education Performance Dashboard** follows a structured methodology to analyze university performance and present meaningful insights through interactive visualizations.

Data Collection

The university ranking dataset was collected from Kaggle. The dataset contains information about universities, including rankings, research performance, student population, internationalization, and country/region details.

Data Preparation

The collected dataset was cleaned and prepared using Python. The process included:

- Loading and inspecting the dataset.
- Identifying missing and duplicate values.
- Removing unnecessary data.
- Cleaning and standardizing column names.
- Correcting data types.
- Creating useful calculated fields such as university age.
- Saving the final cleaned dataset as Cleaned dataset.csv.

KPI Analysis

Key Performance Indicators (KPIs) were created to provide a quick summary of university performance. Important measures include:

- Total Universities
- Average Overall Score
- Average Research Quality
- Total Students
- Average International Students Percentage
- Average Teaching and Industry Scores

Comparative Analysis

Universities were compared based on region, country, university type, research quality, overall score, and student population. This helps identify differences and performance patterns across different groups.

Visualization

The cleaned data was imported into Tableau and converted into interactive dashboards. Different visualization types such as bar charts, pie charts, maps, scatter plots, and tree maps were used depending on the type of analysis.

Interactive Analysis

Filters such as Region, Country/Territory, and Status allow users to explore the data dynamically. When a filter is selected, the related KPIs and visualizations update automatically.

Insight Generation

The final dashboards help users identify high-performing universities, research trends, student population patterns, internationalization levels, and country-wise differences in higher education performance.

Key Insights

The **EduVision DV – Higher Education Performance Dashboard** provides several important insights into university performance, research, and student-related characteristics.

1. **Overall University Performance**

The dashboard helps identify universities with higher overall performance scores and compare their performance across different regions and countries.

2. **Regional Performance Differences**

University performance varies across regions. Comparing average overall scores and research quality helps identify regions with stronger higher-education performance.

3. **Research Performance**

Research quality and research environment provide useful information about the research strength of universities. Universities with stronger research indicators can be identified through the Research Analytics dashboard.

4. **Student Population**

The analysis shows differences in total student population among universities, regions, and university types. This helps identify institutions with larger student populations.

5. **Internationalization**

The percentage of international students and international student diversity provides an understanding of how internationally diverse universities are.

6. **Country-wise Comparison**

The Country Comparison dashboard makes it easier to compare countries based on overall score, research quality, student population, and international student percentage.

7. **Relationship Between Indicators**

Scatter charts help examine relationships between Overall Score and Research Quality and between International Students Percentage and Overall Score, providing a better understanding of how different indicators are related.

8. **Interactive Insights**

The use of Region, Country/Territory, and Status filters allows users to explore specific groups and obtain focused insights without manually analyzing the complete dataset.

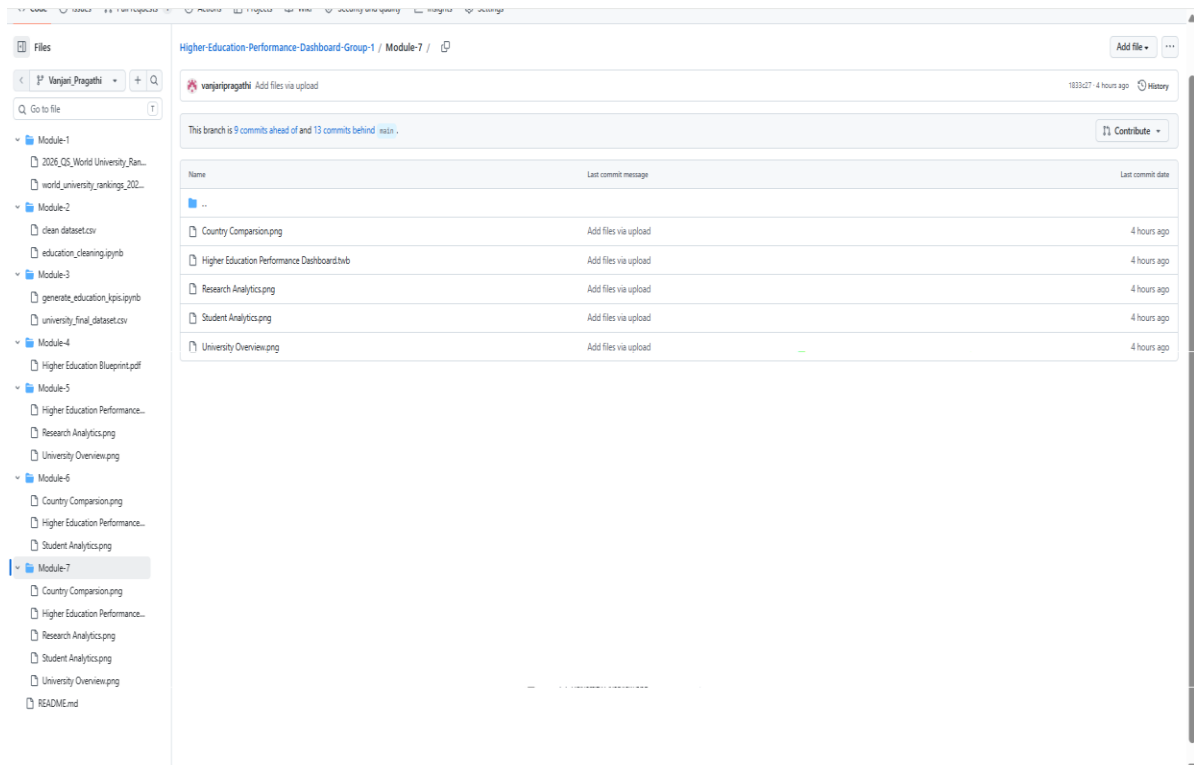
Tools & Technologies

The **EduVision DV – Higher Education Performance Dashboard** was developed using different tools for data collection, preparation, analysis, and visualization.

Tool / Technology	Purpose
Python	Used for data cleaning, preparation, transformation, and handling missing or inconsistent data.
Pandas	Used in Python for loading, analyzing, cleaning, and transforming the university dataset.
Jupyter Notebook	Used as the environment for performing Python-based data preparation and analysis.
Kaggle	Used as the source for collecting the university ranking dataset.
Tableau	Used to create interactive dashboards, KPIs, charts, maps, filters, and visual analytics.
CSV	Used to store and transfer the cleaned university dataset into Tableau.
GitHub	Planned for storing and managing the project files and documentation.
Tableau Public	Can be used to publish and share the completed dashboard online.

Project Folder Structure

The project files are organized into different modules based on the stages of development. The repository contains the original datasets, data-cleaning notebooks, KPI generation files, dashboard screenshots, Tableau workbook, and project documentation.



EduVision DV Project Folder Structure in GitHub

Module	Description
Module-1	Contains the original university ranking datasets.
Module-2	Contains the cleaned dataset and Python/Jupyter notebook used for data cleaning.
Module-3	Contains KPI generation files and the final university dataset.
Module-4	Contains the Higher Education project blueprint.
Module-5	Contains initial dashboard designs and screenshots.
Module-6	Contains dashboard development outputs and screenshots.
Module-7	Contains the final dashboard screenshots and README file.

Deployment

The EduVision DV – Higher Education Performance Dashboard project is organized and maintained in a GitHub repository. The repository contains the project datasets, Python/Jupyter notebooks, KPI generation files, Tableau workbook, dashboard screenshots, and documentation.

GitHub Repository

The complete project is uploaded to GitHub, where the project files are organized into different modules for easy access and management.

GitHub Repository:

[Higher Education Performance Dashboard – Pragathi Branch](#)

Tableau Workbook

The final Tableau workbook is included in the GitHub repository. It contains the four dashboards developed for the project:

- University Overview
- Research Analytics
- Student Analytics
- Country Comparison

Deployment Workflow

Dataset → Python Data Cleaning → KPI Generation → Tableau Dashboard → GitHub Repository

Conclusion

The EduVision DV – Higher Education Performance Dashboard successfully transforms university data into an interactive and user-friendly analytics solution. The project covers important areas such as university performance, research quality, student population, internationalization, and country-wise comparison.

The project involved data collection, cleaning, analysis, KPI creation, and visualization using Python, Pandas, and Tableau. Four interactive dashboards were developed: University Overview, Research Analytics, Student Analytics, and Country Comparison.

Different visualizations such as bar charts, pie charts, maps, treemaps, heat maps, and scatter plots were used to present the data clearly. Filters for Region, Country/Territory, and Status allow users to explore the information according to their requirements.

Overall, the project provides meaningful insights into higher-education performance and demonstrates practical skills in data analysis, visualization, dashboard development, and project documentation. The complete project is organized in GitHub, making it suitable for academic evaluation and portfolio presentation.

Future Scope

The EduVision DV – Higher Education Performance Dashboard can be further improved with additional features and advanced analytics. Future enhancements may include:

- Real-time data updates to keep university information current.
- Predictive analysis to identify future university performance trends.
- Adding more universities, countries, and ranking datasets for wider comparison.
- Developing more advanced interactive visualizations and filters.
- Adding year-wise comparison to analyze changes in university performance over time.
- Using machine learning techniques to predict university rankings or scores.
- Improving the dashboard for mobile and web accessibility.

These improvements can make EduVision DV more powerful, scalable, and useful for students, researchers, institutions, and other users interested in higher-education analysis.